

Study Session 8 — Non-Linear Models & GAMs

Masters-Level DIY Data Science Curriculum (Tips Dataset)

1. Warm-Up & Context (5–10 min)

- ☐ Skim prior sessions: multiple regression, VIF, regularization.
- ☐ Remind yourself why linear models can be too rigid (only straight lines).
- ☐ Write a one-sentence goal for this session (e.g., “Understand what a GAM buys me over OLS”).

2. Load Data & Simple Checks (5–10 min)

- ☐ Load tips dataset (e.g., `tips_cleaned` or `tips_synthetic`) into `df`.
- ☐ Confirm columns: `bill_total_usd`, `party_size`, `tip_usd`, `gender`, `is_smoker`, `tip_pct`.
- ☐ Create `df_work` copy and safely encode `gender_Male` and `is_smoker_Yes` if needed.
- ☐ Run `df_work.describe()` and sanity-check ranges.

3. Non-Linearity & Where Linear Regression Struggles (10–15 min)

- ☐ Plot `tip_usd` vs `bill_total_usd` with a straight regression line (OLS).
- ☐ Inspect residuals vs predicted tips; look for curvature or fanning patterns.
- ☐ Jot down 1–2 spots where you suspect non-linear behavior (e.g., very large bills, large party_size).

4. Baseline Linear Model (Reference Point) (10–15 min)

- ☐ Fit a multiple linear regression: `tip_usd ~ bill_total_usd + party_size + gender_Male + is_smoker_Yes`.
- ☐ Record R^2 , Adjusted R^2 , and RMSE (train and/or test).
- ☐ Save ‘Actual vs Predicted Tip’ plot with `save_and_log_plot()`.
- ☐ Save residuals vs predicted plot to compare later against GAM.

5. GAM Concept — Flexible Curves (10–15 min)

- ☐ Write down the GAM idea in words: ‘each feature gets its own smooth curve.’
- ☐ Note the additive form: $\text{prediction} = \text{baseline} + f_1(x_1) + f_2(x_2) + \dots$
- ☐ List which features you’ll model with smooths (`bill_total_usd`, `party_size`, maybe `gender_Male`, `is_smoker_Yes`).
- ☐ In Markdown, summarize when you’d choose a GAM over plain OLS.

6. Fit a GAM in Python (20–25 min)

- ☐ Install/import `pygam` (or chosen GAM library).

- ☐ Define feature matrix X and target y (match your earlier linear model).
- ☐ Fit LinearGAM with one smooth term per feature (e.g., $s(0) + s(1) + \dots$).
- ☐ Print/inspect `model.summary()` just to see overall behavior (don't obsess over p-values).

7. Plot & Interpret GAM Smooths (20–25 min)

- ☐ Create one plot per feature showing: x-axis = feature, y-axis = smooth effect on `tip_usd`.
- ☐ Add a horizontal line at 0 to show 'no effect' baseline.
- ☐ Save the combined 'GAM smooths' plot using `save_and_log_plot()`.
- ☐ In Markdown, ELI5 each smooth: what story does the curve tell?

8. Compare OLS vs GAM (15–20 min)

- ☐ Compute R^2 and RMSE for both OLS and GAM on the same train/test split.
- ☐ Make a simple side-by-side table: model, R^2 , RMSE.
- ☐ Write 2–3 bullet points: did GAM meaningfully improve fit?
- ☐ Note where GAM helps most (e.g., high bills, large parties, odd corners of the data).

9. Diagnostics & Overfitting Check (Optional, 10–15 min)

- ☐ Visually compare residual plots for OLS vs GAM.
- ☐ Check if GAM overfits: does test R^2 drop vs train?
- ☐ If overfitting appears, try adjusting GAM regularization or number of splines.

10. Reflection (5–10 min)

- ☐ In your own words: What can a GAM do that linear regression can't?
- ☐ Write a one-sentence rule of thumb for when you'd reach for a GAM.
- ☐ Note 1–2 real-world problems (outside tipping) where GAMs might shine.

Tip: Next time you see curved residuals or fanning patterns, consider a GAM as a flexible, interpretable upgrade to your linear model.